

HUNDRED DAYS OF CODING C PROGRAMMING

DAY :- 11

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BATCH :- 15

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Question : 1

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. Find the sum of squares of its digits using a while loop. E.g. Number: 123, Sum=14

Step 1 — Algorithm

Algorithm: Menu-Driven Program for Digit Operations

Start

Enter a positive integer N.

Display the following menu:

Sum of squares of digits

Sum of cubes of digits

Difference between sum of cubes and sum of squares

Enter the user's choice.

Use switch-case according to the choice.

If choice is 1, find the sum of squares of digits using a while loop.

If choice is 2, find the sum of cubes of digits using a for loop.

If choice is 3, find the sum of squares and sum of cubes using a do-while loop, then calculate their difference.

Display the result.

If the choice is invalid, display "Invalid Choice".

Stop.

Example: For N = 123

Sum of squares = $1^2 + 2^2 + 3^2 = 14$

Sum of cubes = $1^3 + 2^3 + 3^3 = 36$

Difference = $36 - 14 = 22$

Step 2 — Test Case Table

TEST CASE	INPUT NUMBER	CHOICE	EXPECTED OUTPUT
TC1	123	1	Sum of squares = 14
TC2	123	2	Sum of cubes = 36
TC3	123	3	Difference = 22
TC4	45	1	Sum of squares = 41
TC5	45	2	Sum of cubes = 189
TC6	45	3	Difference = 148
TC7	7	1	Sum of squares = 49
TC8	7	2	Sum of cubes = 343
TC9	7	3	Difference = 294
TC10	123	4	Invalid choice

Step 3 — C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, choice, digit;    int squareSum = 0,  
    cubeSum = 0, difference;
```

```
printf("Enter a positive integer: ");  
scanf("%d", &n);
```

```
printf("\n1. Sum of squares of digits"); printf("\n2. Sum of cubes  
of digits"); printf("\n3. Difference between sum of cubes and sum  
of squares");
```

```
printf("\nEnter your choice: ");  
scanf("%d", &choice);
```

```
switch(choice)  
{ case 1:  
while(n > 0)  
{  
    digit = n % 10;    squareSum =  
squareSum + digit * digit;  
    n = n / 10;  
}
```

```
printf("Sum of squares of digits = %d", squareSum);  
break;
```

case 2:

```
while(n > 0)
{
    digit = n % 10;      cubeSum =
cubeSum + digit * digit * digit;
    n = n / 10;
}

printf("Sum of cubes of digits = %d", cubeSum);
break;
```

case 3:

```
{
    int temp = n;

do
{
    digit = temp % 10;      squareSum =
squareSum + digit * digit;    cubeSum =
cubeSum + digit * digit * digit;    temp =
temp / 10;
}
while(temp > 0);

difference = cubeSum - squareSum;
```

```
        printf("Difference = %d", difference);  
break;  
    }  
  
    default:  
        printf("Invalid Choice");  
    }  
  
    return 0;  
}
```

Step 4 — Compile the Program

```
1 #include <stdio.h>
2 int main()
3 {
4     int n, choice, digit;
5     int squareSum = 0, cubeSum = 0, difference;
6     printf("Enter a positive integer: ");
7     scanf("%d", &n);
8     printf("\n1. Sum of squares of digits");
9     printf("\n2. Sum of cubes of digits");
10    printf("\n3. Difference between sum of cubes and sum of squares");
11    printf("\nEnter your choice: ");
12    scanf("%d", &choice);
13    switch(choice)
14    {
15        case 1:
16            while(n > 0)
17            {
18                digit = n % 10;
19                squareSum = squareSum + digit * digit;
20                n = n / 10;
21            }
22            printf("Sum of squares of digits = %d", squareSum);
23            break;
24        case 2:
25            while(n > 0)
26            {
27                digit = n % 10;
28                cubeSum = cubeSum + digit * digit * digit;
29                n = n / 10;
30            }
31            printf("Sum of cubes of digits = %d", cubeSum);
32            break;
33        case 3:
34            {
35                int temp = n;
36                do
37                {
38                    digit = temp % 10;
39                    squareSum = squareSum + digit * digit;
40                    cubeSum = cubeSum + digit * digit * digit;
41                    temp = temp / 10;
42                }
43                while(temp > 0);
44                difference = cubeSum - squareSum;
45                printf("Difference = %d", difference);
46                break;
47            }
48        default:
49            printf("Invalid Choice");
50    }
51    return 0;
52 }
```

Step 5 — Execute the Program and Obtain the Output

```
1 #include <stdio.h>
2 int main()
3 {
4     int n, choice, digit;
5     int squareSum = 0, cubeSum = 0, difference;
6     printf("Enter a positive integer: ");
7     scanf("%d", &n);
8     printf("\n1. Sum of squares of digits");
9     printf("\n2. Sum of cubes of digits");
10    printf("\n3. Difference between sum of cubes and sum of squares");
11    printf("\nEnter your choice: ");
12    scanf("%d", &choice);
13    switch(choice)
14    {
15        case 1:
16            while(n > 0)
17            {
18                digit = n % 10;
19                squareSum = squareSum + digit * digit;
20                n = n / 10;
21            }
22            printf("Sum of squares of digits = %d", squareSum);
23            break;
24        case 2:
25            while(n > 0)
26            {
27                digit = n % 10;
28                cubeSum = cubeSum + digit * digit * digit;
29                n = n / 10;
30            }
31            printf("Sum of cubes of digits = %d", cubeSum);
32            break;
33        case 3:
34            {
35                int temp = n;
36                do
37                {
38                    digit = temp % 10;
39                    squareSum = squareSum + digit * digit;
40                    cubeSum = cubeSum + digit * digit * digit;
41                    temp = temp / 10;
42                }
43                while(temp > 0);
44                difference = cubeSum - squareSum;
45                printf("Difference = %d", difference);
46                break;
47            }
48        default:
49            printf("Invalid Choice");
50    }
51    return 0;
52 }
```

```
Enter a positive integer: 123

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 2
Sum of cubes of digits = 36

...Program finished with exit code 0
Press ENTER to exit console.
```

Step 6 — Test the Program with Prepared Test Cases

Case 1

```
Enter a positive integer: 123

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 1
Sum of squares of digits = 14

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 2

```
Enter a positive integer: 123

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 2
Sum of cubes of digits = 36

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 3

```
Enter a positive integer: 123

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 3
Difference = 22

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 4

```
Enter a positive integer: 45

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 1
Sum of squares of digits = 41

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 5


```
Enter a positive integer: 45

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 2
Sum of cubes of digits = 189

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 6

```
Enter a positive integer: 45

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 3
Difference = 148

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 7

```
Enter a positive integer: 7

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 1
Sum of squares of digits = 49

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 8

```
Enter a positive integer: 7

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 2
Sum of cubes of digits = 343

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 9

```
Enter a positive integer: 7

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 3
Difference = 294

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 10

```
Enter a positive integer: 123

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 4
Invalid Choice

...Program finished with exit code 0
Press ENTER to exit console.
```

Step 7 — Re-verify the Program Against the Test Cases

The program was re-verified using the prepared test cases, and the actual output was found to be the same as the expected output.

Result: All test cases passed successfully.

Step - 8

Conclusion: The menu-driven C program was successfully written, compiled, executed, tested, and re-verified. All the test cases gave the expected results.

Question 2

Find the sum of cubes of its digits using a for loop. E.g. Number: 123, Sum= 36

Step 1 — Algorithm

- 1.Start
- 2.Enter a positive integer N.
- 3.Display the menu of three operations.
- 4.Enter the user's choice.
- 5.Use switch-case according to the choice.
- 6.If choice is 1, find the sum of squares of digits using a while loop.

7.If choice is 2, find the sum of cubes of digits using a for loop.

8.If choice is 3, find the difference between the sum of cubes and the sum of squares using a

9.do-while loop.

Display the result.

10.For an invalid choice, display "Invalid Choice".

11.Stop.

Step 2 — Test Case Table

TEST CASE	INPUT NUMBER	EXPECTED OUTPUT
TC1	123	Sum of Cubes = 36
TC2	45	Sum of Cubes = 189
TC3	7	Sum of Cubes = 343
TC4	12	Sum of Cubes = 9
TC5	25	Sum of Cubes = 133

Step 3 — C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, digit, sum = 0;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%d", &n);
```

```
    for (; n > 0; n = n / 10)
```

```
    {
```

```
        digit = n % 10;    sum = sum +
```

```
        digit * digit * digit;
```

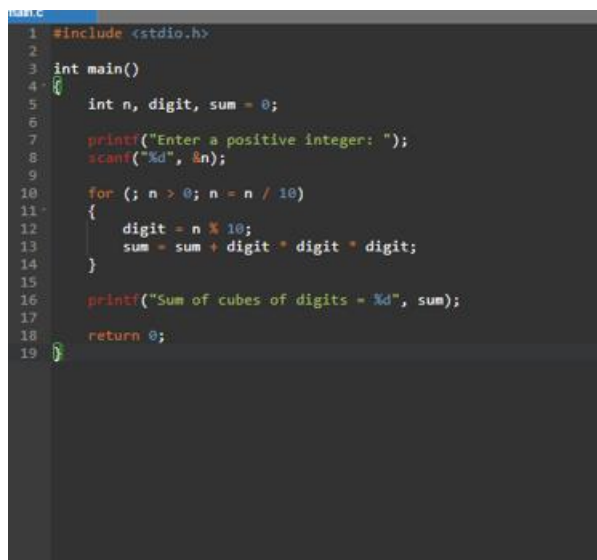
```
}
```

```
printf("Sum of cubes of digits = %d", sum);
```

```
return 0;
```

```
}
```

Step 4 — Compile the C Program

A screenshot of a code editor showing a C program. The code is as follows:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, digit, sum = 0;
6
7     printf("Enter a positive integer: ");
8     scanf("%d", &n);
9
10    for (; n > 0; n = n / 10)
11    {
12        digit = n % 10;
13        sum = sum + digit * digit * digit;
14    }
15
16    printf("Sum of cubes of digits = %d", sum);
17
18    return 0;
19 }
```

The code is color-coded: keywords like `include`, `int`, `main`, `for`, `return`, and `printf` are in blue; variables like `n`, `digit`, and `sum` are in green; string literals are in red; and operators and punctuation are in white. The editor has a dark background and a light-colored border.

Step 5 — Execute the Program and Obtain the Output

```
main.c
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, digit, sum = 0;
6
7      printf("Enter a positive integer: ");
8      scanf("%d", &n);
9
10     for (; n > 0; n = n / 10)
11     {
12         digit = n % 10;
13         sum = sum + digit * digit * digit;
14     }
15
16     printf("Sum of cubes of digits = %d", sum);
17
18     return 0;
19 }
```

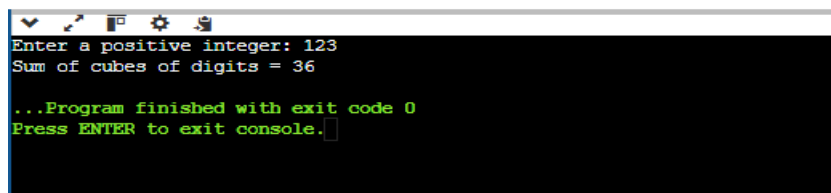
Enter a positive integer: 123
Sum of cubes of digits = 36

...Program finished with exit code 0
Press ENTER to exit console.

Step 6 — Test the Program with Prepared Test Cases

Case 1

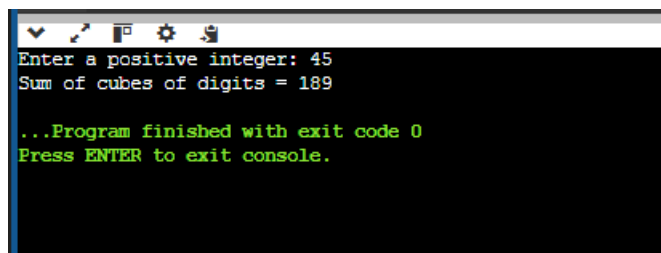
.



```
Enter a positive integer: 123
Sum of cubes of digits = 36

...Program finished with exit code 0
Press ENTER to exit console.
```

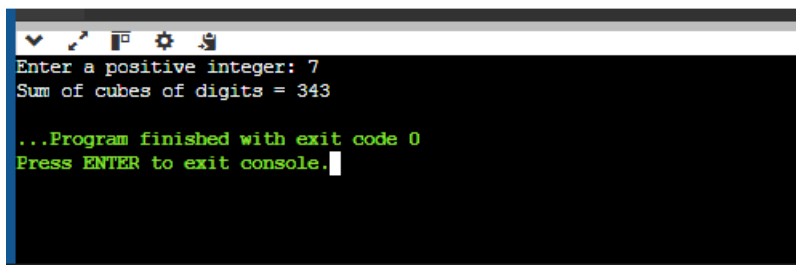
Case 2



```
Enter a positive integer: 45
Sum of cubes of digits = 189

...Program finished with exit code 0
Press ENTER to exit console.
```

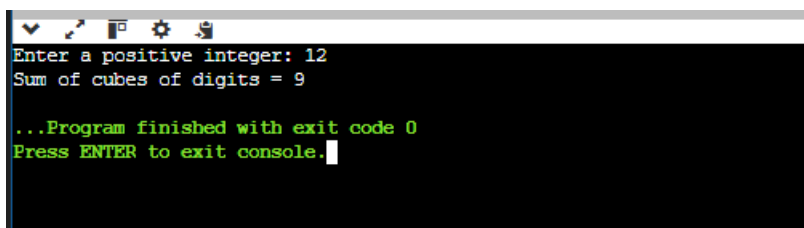
Case 3

A screenshot of a console window with a dark background and a light gray title bar. The title bar contains standard window control icons. The console text is as follows:

```
Enter a positive integer: 7
Sum of cubes of digits = 343

...Program finished with exit code 0
Press ENTER to exit console.
```

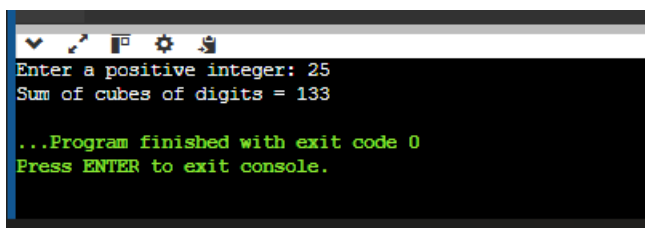
Case 4

A screenshot of a console window with a dark background and a light gray title bar. The title bar contains standard window control icons. The console text is as follows:

```
Enter a positive integer: 12
Sum of cubes of digits = 9

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 5

A screenshot of a console window with a dark background and a light gray title bar. The title bar contains standard window control icons. The console text is as follows:

```
Enter a positive integer: 25
Sum of cubes of digits = 133

...Program finished with exit code 0
Press ENTER to exit console.
```

Step 7 — Re-verify the Program Against the Test Cases

The program was re-verified using the prepared test cases. All test cases passed successfully and the actual outputs matched the expected outputs.

Step 8 — Conclusion

The C program to find the sum of cubes of the digits using a for loop was successfully written, compiled, executed, tested, and re-verified. All the test cases gave the expected results.

Question 3

Find the sum of cubes of its digits using a for loop. E.g. Number: 123, Sum= 36

Step 1 — Algorithm

- 1.Start
- 2.Enter a positive integer N.
- 3.Set squareSum = 0 and cubeSum = 0.
- 4.Extract the last digit of N.
- 5.Add the square of the digit to squareSum.
- 6.Add the cube of the digit to cubeSum.
- 7.Remove the last digit from N.
- 8.Repeat steps 4–7 using a do-while loop until all digits are processed.
- 9.Calculate: Difference = cubeSum – squareSum
- 10.Display the difference.
- 11.Stop.

Step 2 — Test Case Table

Test Case	Input Number	Expected Output
TC1	123	Difference :- 22
TC2	45	Difference :- 148
TC3	7	Difference :- 294
TC4	12	Difference :- 6
TC5	25	Difference :- 108

Step 3 — C Program

```
#include <stdio.h>

int main()
{
    int n,
    digit;

    int squareSum = 0, cubeSum = 0, difference;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    do
    {
        digit = n % 10;

        squareSum = squareSum + digit * digit;
        cubeSum = cubeSum + digit * digit * digit;

        n = n / 10;

    } while(n > 0);

    difference = cubeSum - squareSum;

    printf("Sum of squares = %d\n", squareSum);
    printf("Sum of cubes = %d\n", cubeSum);
    printf("Difference = %d", difference);
```

```
    return 0;
}
```

Step 4 — Compile the C Program

```
main.c
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, digit;
6      int squareSum = 0, cubeSum = 0, difference;
7
8      printf("Enter a positive integer: ");
9      scanf("%d", &n);
10
11     do
12     {
13         digit = n % 10;
14
15         squareSum = squareSum + digit * digit;
16         cubeSum = cubeSum + digit * digit * digit;
17
18         n = n / 10;
19
20     } while(n > 0);
21
22     difference = cubeSum - squareSum;
23
24     printf("Sum of squares = %d\n", squareSum);
25     printf("Sum of cubes = %d\n", cubeSum);
26     printf("Difference = %d", difference);
27
28     return 0;
29 }
30
```

Step 5 — Execute the Program and Obtain the Output

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, digit;
6      int squareSum = 0, cubeSum = 0, difference;
7
8      printf("Enter a positive integer: ");
9      scanf("%d", &n);
10
11     do
12     {
13         digit = n % 10;
14
15         squareSum = squareSum + digit * digit;
16         cubeSum = cubeSum + digit * digit * digit;
17
18         n = n / 10;
19     } while(n > 0);
20
21     difference = cubeSum - squareSum;
22
23     printf("Sum of squares = %d\n", squareSum);
24     printf("Sum of cubes = %d\n", cubeSum);
25     printf("Difference = %d", difference);
26
27     return 0;
28 }
```

Enter a positive integer: 45
Sum of squares = 41
Sum of cubes = 189
Difference = 148

...Program finished with exit code 0
Press ENTER to exit console.

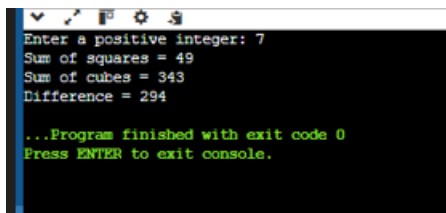
Step 6 — Test the Program with Prepared Test Cases

Case 1

```
Enter a positive integer: 45  
Sum of squares = 41  
Sum of cubes = 189  
Difference = 148
```

...Program finished with exit code 0
Press ENTER to exit console.

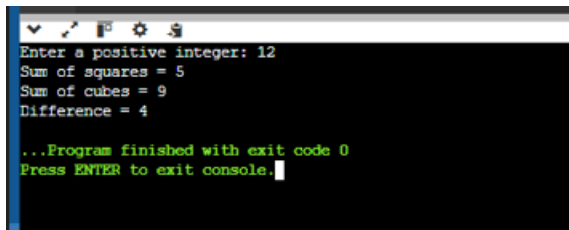
Case 2



```
Enter a positive integer: 7
Sum of squares = 49
Sum of cubes = 343
Difference = 294

...Program finished with exit code 0
Press ENTER to exit console.
```

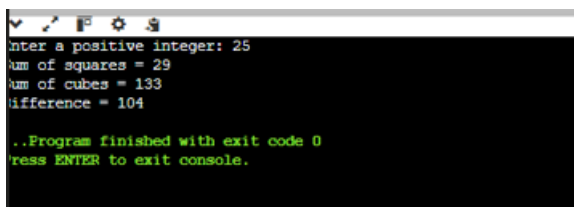
Case 3



```
Enter a positive integer: 12
Sum of squares = 5
Sum of cubes = 9
Difference = 4

...Program finished with exit code 0
Press ENTER to exit console.
```

Case 4



```
Enter a positive integer: 25
Sum of squares = 29
Sum of cubes = 133
Difference = 104

...Program finished with exit code 0
Press ENTER to exit console.
```

Step 7 — Re-verify the Program Against the Test Cases

The program was re-verified using the prepared test cases. All test cases passed successfully and the actual outputs matched the expected outputs.

Step 8 — Conclusion

The C program to find the difference between the sum of cubes and the sum of squares of digits using a do-while loop was successfully written, compiled, executed, tested, and reverified. All the test cases gave the expected results.